Experiment No 7

Vulnerability Analysis Using CGI Scanning with Nikto

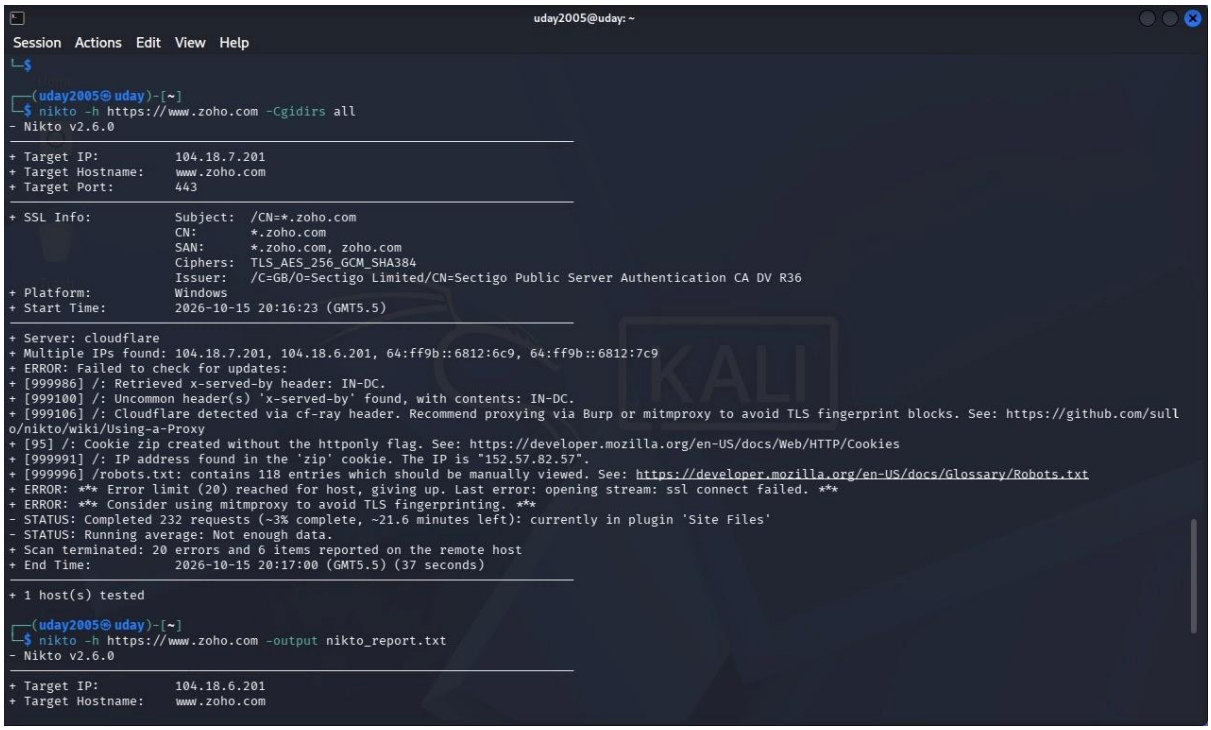
Aim

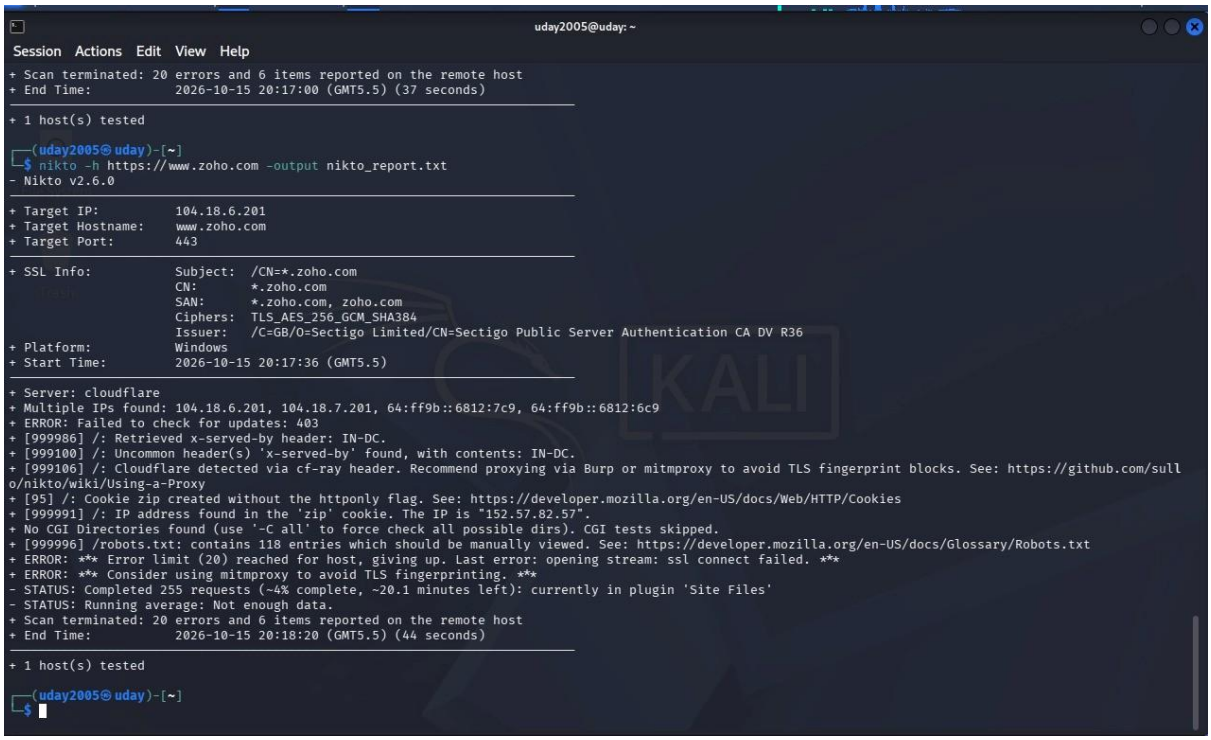
To analyze web server vulnerabilities using the Nikto tool and identify potentially vulnerable CGI directories on a target web server

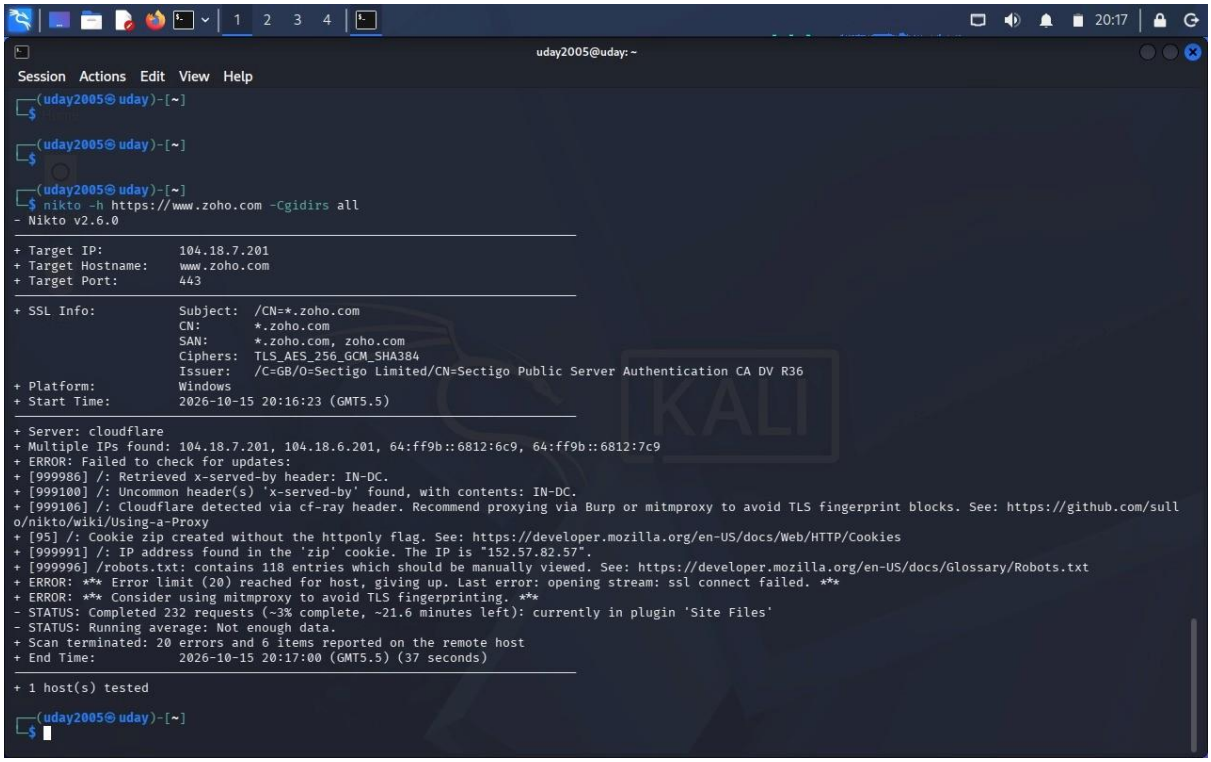
Algorithm

1. Start Kali Linux and open the terminal.
2. Check the Nikto help menu using nikto -H.
3. Specify the authorized target website using the Nikto -h option.
4. Perform a web server scan using nikto -h https://www.zoho.com.
5. Scan for CGI directories using the -Cgidirs all option on an authorized target.
6. Observe the scan output and identify reported directories, HTTP status codes, and potential security issues.
7. Record the findings and stop the scan.

Output




uday2005@uday: ~
Session Actions Edit View Help
+ Scan terminated: 20 errors and 6 items reported on the remote host
+ End Time: 2026-10-15 20:17:00 (GMT5.5) (37 seconds)
+ 1 host(s) tested
(uday2005㉿uday)-[~]
$ nikto -h https://www.zoho.com -output nikto_report.txt
- Nikto v2.6.0
+ Target IP: 104.18.6.201
+ Target Hostname: www.zoho.com
+ Target Port: 443
+ SSL Info: Subject: /CN=*.zoho.com
CN: *.zoho.com
SAN: *.zoho.com, zoho.com
Ciphers: TLS_AES_256_GCM_SHA384
Issuer: /C=GB/O=Sectigo Limited/CN=Sectigo Public Server Authentication CA DV R36
+ Platform: Windows
+ Start Time: 2026-10-15 20:17:36 (GMT5.5)
+ Server: cloudflare
+ Multiple IPs found: 104.18.6.201, 104.18.7.201, 64:ff9b::6812:7c9, 64:ff9b::6812:6c9
+ ERROR: Failed to check for updates: 403
+ [999986] /: Retrieved x-served-by header: IN-DC.
+ [999100] /: Uncommon header(s) 'x-served-by' found, with contents: IN-DC.
+ [999106] /: Cloudflare detected via cf-ray header. Recommend proxying via Burp or mitmproxy to avoid TLS fingerprint blocks. See: https://github.com/sullo/nikto/wiki/Using-a-Proxy
+ [95] /: Cookie zip created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
+ [999991] /: IP address found in the 'zip' cookie. The IP is "152.57.82.57".
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [999996] /robots.txt: contains 118 entries which should be manually viewed. See: https://developer.mozilla.org/en-US/docs/Glossary/Robots.txt
+ ERROR: *** Error limit (20) reached for host, giving up. Last error: opening stream: ssl connect failed. ***
+ ERROR: *** Consider using mitmproxy to avoid TLS fingerprinting. ***
- STATUS: Completed 255 requests (~4% complete, ~20.1 minutes left): currently in plugin 'Site Files'
- STATUS: Running average: Not enough data.
+ Scan terminated: 20 errors and 6 items reported on the remote host
+ End Time: 2026-10-15 20:18:20 (GMT5.5) (44 seconds)
+ 1 host(s) tested
(uday2005㉿uday)-[~]
$


1 2 3 4
20:17
uday2005@uday: ~
Session Actions Edit View Help
(uday2005㉿uday)-[~]
$
(uday2005㉿uday)-[~]
$
(uday2005㉿uday)-[~]
$ nikto -h https://www.zoho.com -Cgidirs all
- Nikto v2.6.0
+ Target IP: 104.18.7.201
+ Target Hostname: www.zoho.com
+ Target Port: 443
+ SSL Info: Subject: /CN=*.zoho.com
CN: *.zoho.com
SAN: *.zoho.com, zoho.com
Ciphers: TLS_AES_256_GCM_SHA384
Issuer: /C=GB/O=Sectigo Limited/CN=Sectigo Public Server Authentication CA DV R36
+ Platform: Windows
+ Start Time: 2026-10-15 20:16:23 (GMT5.5)
+ Server: cloudflare
+ Multiple IPs found: 104.18.7.201, 104.18.6.201, 64:ff9b::6812:6c9, 64:ff9b::6812:7c9
+ ERROR: Failed to check for updates:
+ [999986] /: Retrieved x-served-by header: IN-DC.
+ [999100] /: Uncommon header(s) 'x-served-by' found, with contents: IN-DC.
+ [999106] /: Cloudflare detected via cf-ray header. Recommend proxying via Burp or mitmproxy to avoid TLS fingerprint blocks. See: https://github.com/sullo/nikto/wiki/Using-a-Proxy
+ [95] /: Cookie zip created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
+ [999991] /: IP address found in the 'zip' cookie. The IP is "152.57.82.57".
+ [999996] /robots.txt: contains 118 entries which should be manually viewed. See: https://developer.mozilla.org/en-US/docs/Glossary/Robots.txt
+ ERROR: *** Error limit (20) reached for host, giving up. Last error: opening stream: ssl connect failed. ***
+ ERROR: *** Consider using mitmproxy to avoid TLS fingerprinting. ***
- STATUS: Completed 232 requests (~3% complete, ~21.6 minutes left): currently in plugin 'Site Files'
- STATUS: Running average: Not enough data.
+ Scan terminated: 20 errors and 6 items reported on the remote host
+ End Time: 2026-10-15 20:17:00 (GMT5.5) (37 seconds)
+ 1 host(s) tested
(uday2005㉿uday)-[~]
$

Result

The Nikto tool was used to analyze the target web server for potential vulnerabilities and CGI directory exposure. The scan findings can be reviewed to identify possible security weaknesses. Actual vulnerabilities must be confirmed from the scan output.